

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		Conservation of energy or implied (1) Full substitution or good algebra e.g. $\frac{1}{2}mv_b^2 = mgh + \frac{1}{2}mv^2$ OR Initial KE = 0.17 [J] and PE loss = 0.47 [J] (implied by final KE = 0.64 J) OR $v_b = \sqrt{2gh + v^2}$ (1) Final velocity = 6.53 [m s⁻¹] (1) Alternative: Use of $v^2 = u^2 + 2ax$ with $u = 3.35$ m s⁻¹ and $a = 9.81$ m s⁻² (1) Correct answer = 6.53 [m s⁻¹] (1) Some statement of why this equation works e.g. due to conservation of energy, we can consider a particle dropping with initial downward speed of 3.35 (1) Accept – this equation shouldn't work but it does! Alternative: Use of $v^2 = u^2 + 2ax$ with $u = 0$ gives $v = 5.6$ [m s⁻¹] (1) Pythagoras applied i.e. $\sqrt{3.35^2 + 5.6^2}$ (1) Correct answer = 6.53 [m s⁻¹] (1)		3		3	2	
		(ii)		Use of $x = ut + \frac{1}{2}at^2$ for the vertical motion or 2 other usable equations (1) $t = \sqrt{\frac{2x}{g}} = \sqrt{\frac{2(1.2+0.8)}{9.81}} = 0.64$ [s] (1) For the horizontal motion: horizontal distance from point of released = $vt = 3.35 \times 0.64$ ecf (1) = 2.14 [m] so claim is correct (1) ecf Accept $2.14 \approx 2$			4	4	3	
				Question 2 total	4	7	4	15	10	0